

Communicating with Data – Color

Dr. Ab Mosca (they/them)

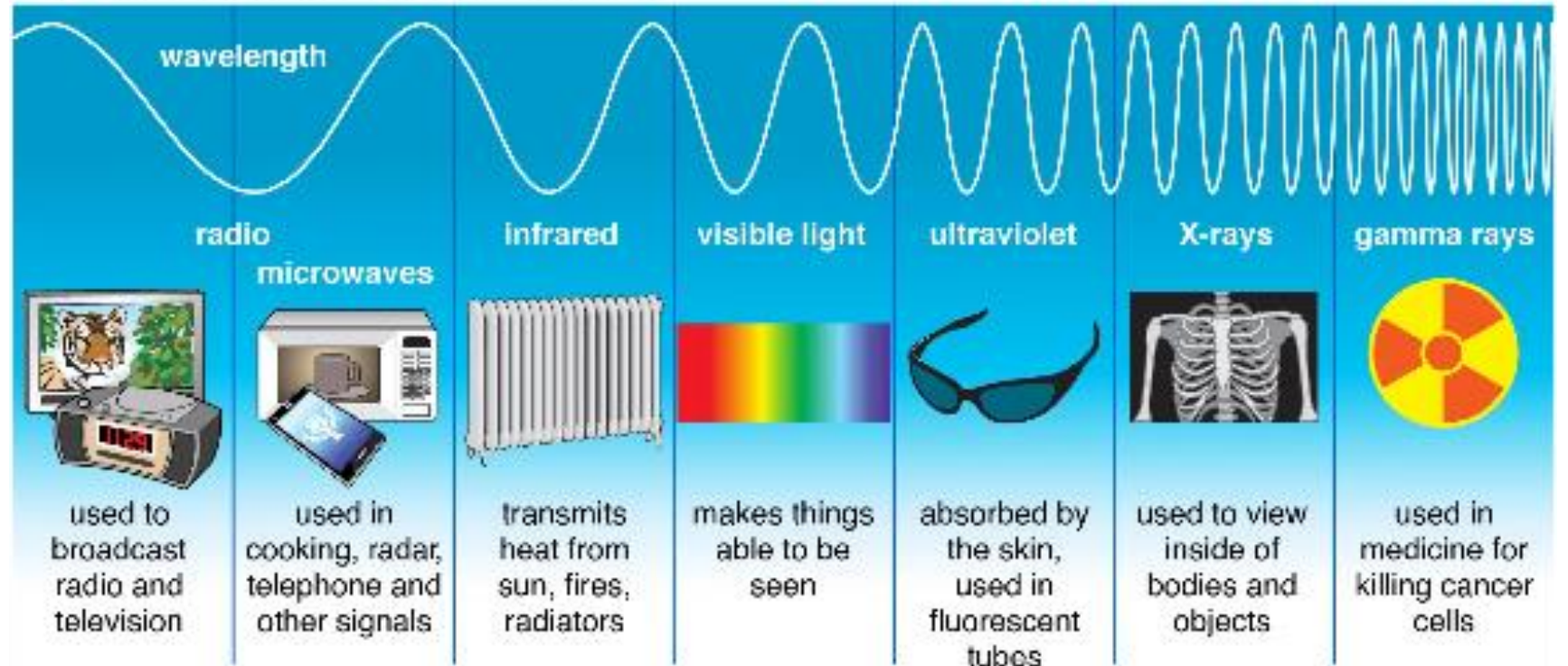
Slides based off slides courtesy of Jordan Crouser (<https://jcrouser.github.io/>)

Plan for Today

- Electromagnetic radiation
- How we see color
- Color for visual encoding
- More perceptual problems

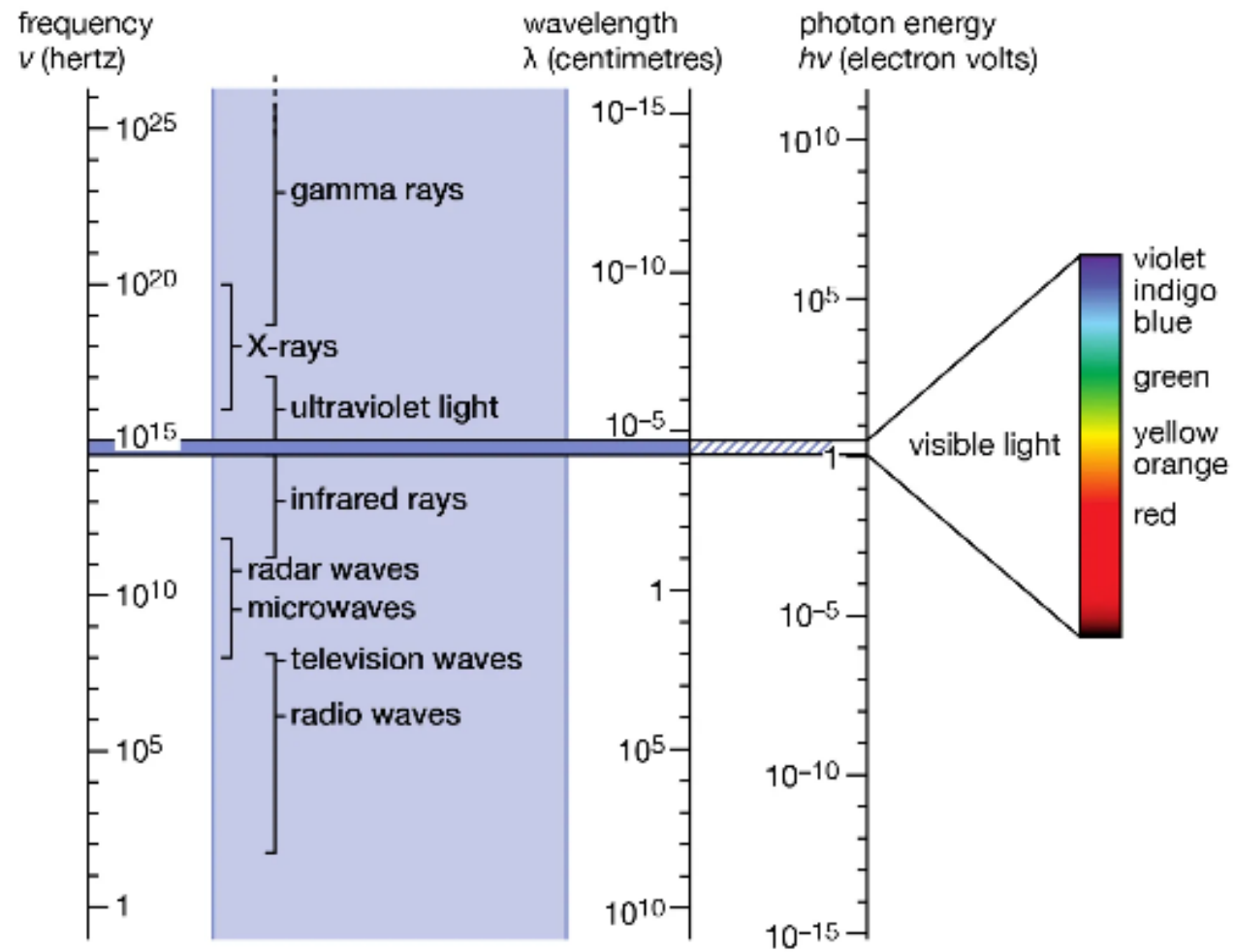
Electromagnetic Radiation

Types of Electromagnetic Radiation



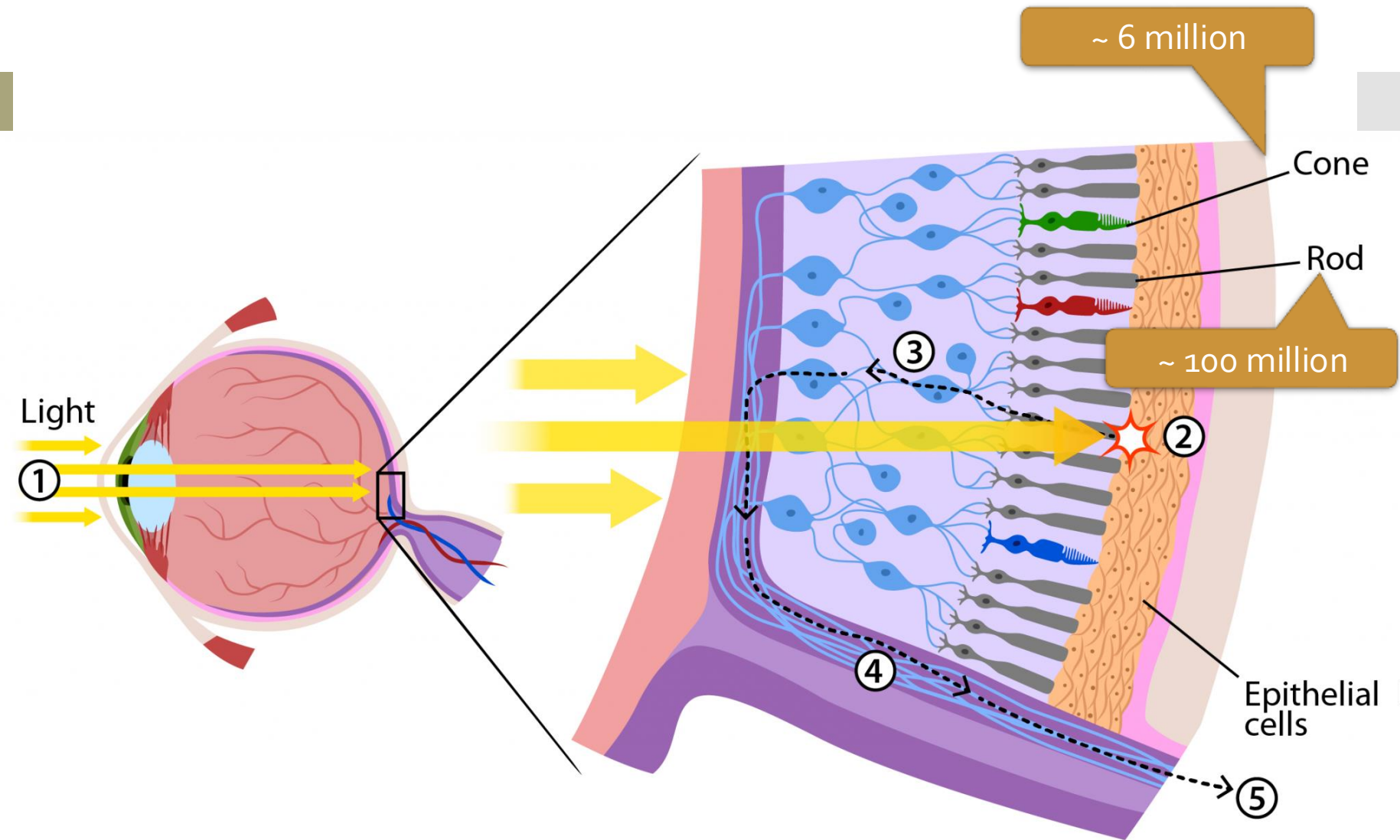
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Visible Light



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How we see color



Recall

What is expressiveness of a visual channel?

What is effectiveness of a visual channel?

Color for Visual Encoding

If we use color to represent our data, an important question to ask is: "How can we choose effective colors that provide good differentiation between data elements during the visualization task?"

C. G. Healey, "Choosing effective colours for data visualization," *Proceedings of Seventh Annual IEEE Visualization '96*, San Francisco, CA, USA, 1996, pp. 263-270, doi: 10.1109/VISUAL.1996.568118.

Color for Visual Encoding

If we use color to represent our data, an important question to ask is: “How can we choose effective colors that provide good differentiation between data elements during the visualization task?”

- How can we allow rapid and accurate identification of individual data elements through the use of color?
- What factors determine whether a “target” element’s color will make it easy to find, relative to differently colored “nontarget” elements?
- How many colors can we display at once, while still allowing for rapid and accurate target identification?

C. G. Healey, "Choosing effective colours for data visualization," *Proceedings of Seventh Annual IEEE Visualization '96*, San Francisco, CA, USA, 1996, pp. 263-270, doi: 10.1109/VISUAL.1996.568118.

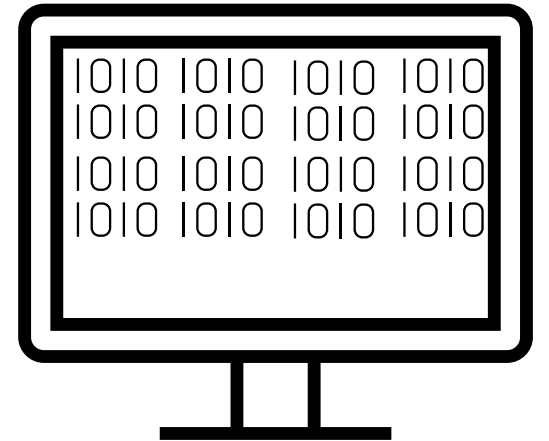
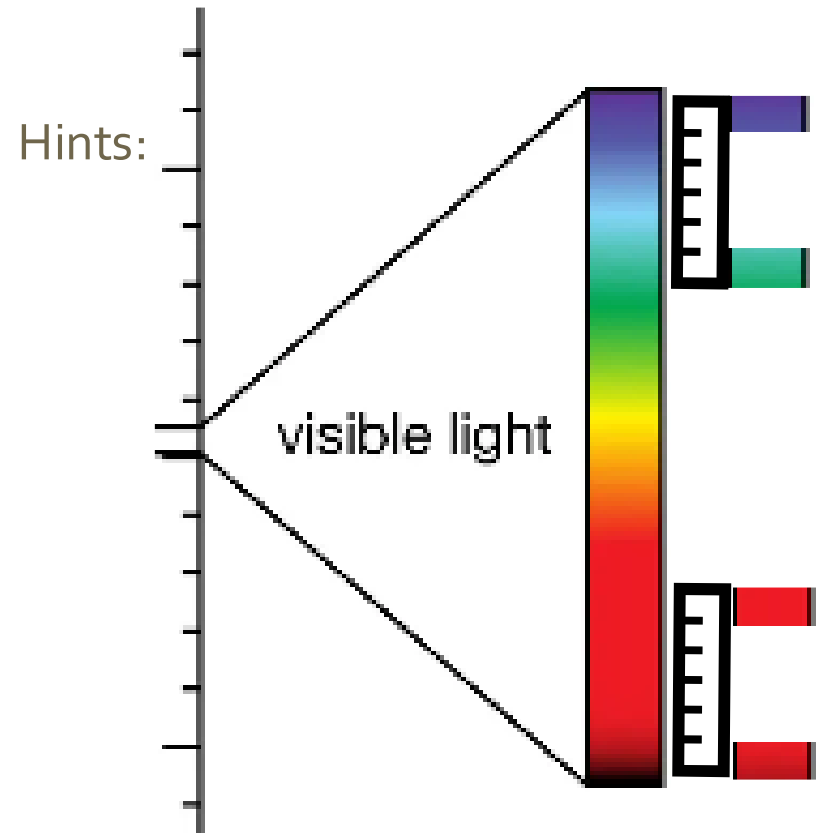
Colormaps

- Colormaps = predetermined sequences of color for representing data values
- 3 Main Principles:
 - **Order:** a good colormap sequence should be naturally orderable (e.g., from a cool blue color to warm red)
 - **Continuity:** the colormap should only reflect actual differences in the data without creating artificial boundaries
 - **Perceptual Uniformity:** adjacent colors should reflect even perceptual distances throughout the sequence, such that a step in data magnitude is matched by an equivalent perceptual step in color.

Colormaps

Perceptual Uniformity: adjacent colors should reflect even perceptual distances throughout the sequence, such that a step in data magnitude is matched by an equivalent perceptual step in color.

Why is this challenging?



Color for Visual Encoding

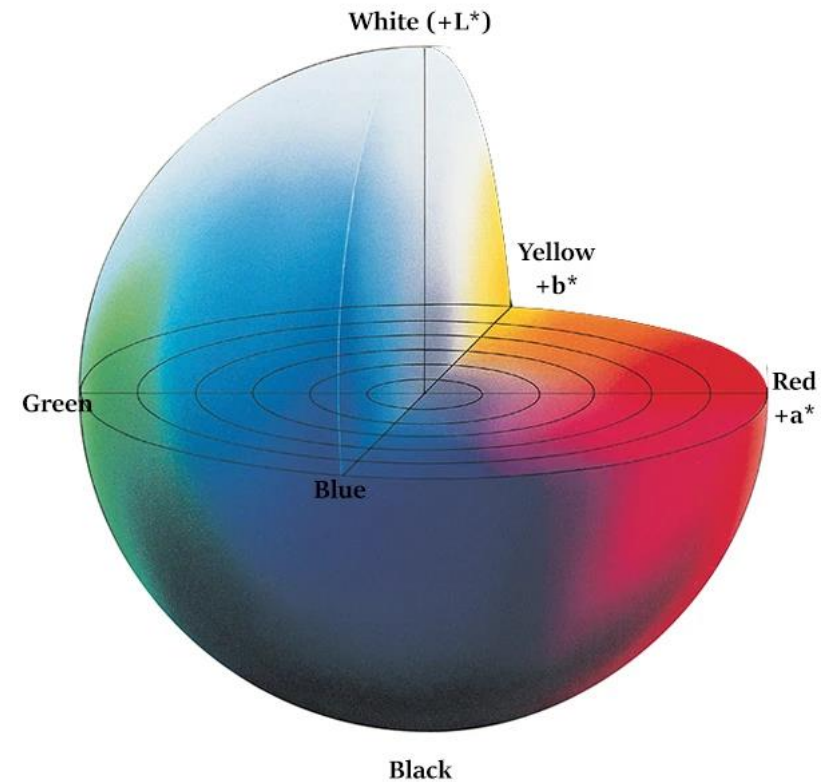
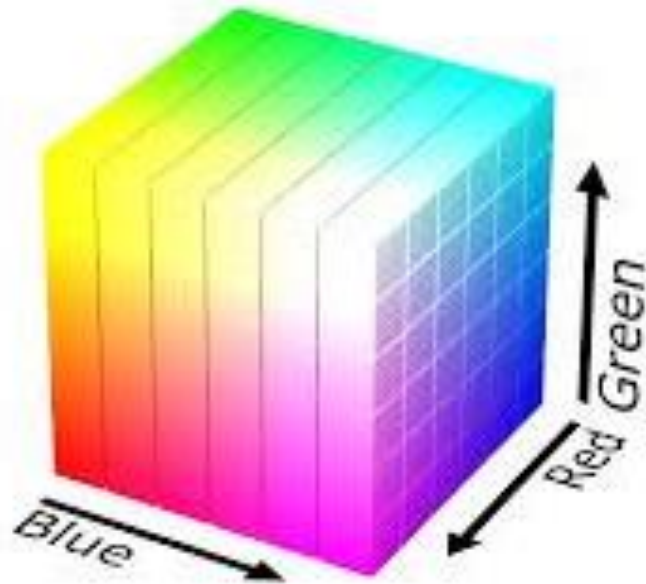
One approach—Control:

1. Color Distance
2. Linear separation of colors
3. Color category similarity

C. G. Healey, "Choosing effective colours for data visualization," *Proceedings of Seventh Annual IEEE Visualization '96*, San Francisco, CA, USA, 1996, pp. 263-270, doi: 10.1109/VISUAL.1996.568118.

Color for Visual Encoding

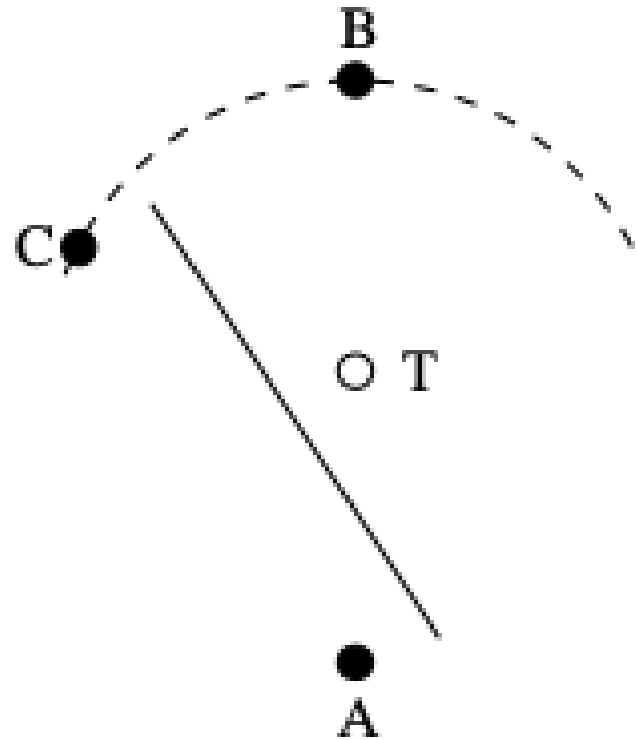
Color Distance: the Euclidean distance between different colors as measured in a perceptually balanced color model



C. G. Healey, "Choosing effective colours for data visualization," *Proceedings of Seventh Annual IEEE Visualization '96*, San Francisco, CA, USA, 1996, pp. 263-270, doi: 10.1109/VISUAL.1996.568118.

Color for Visual Encoding

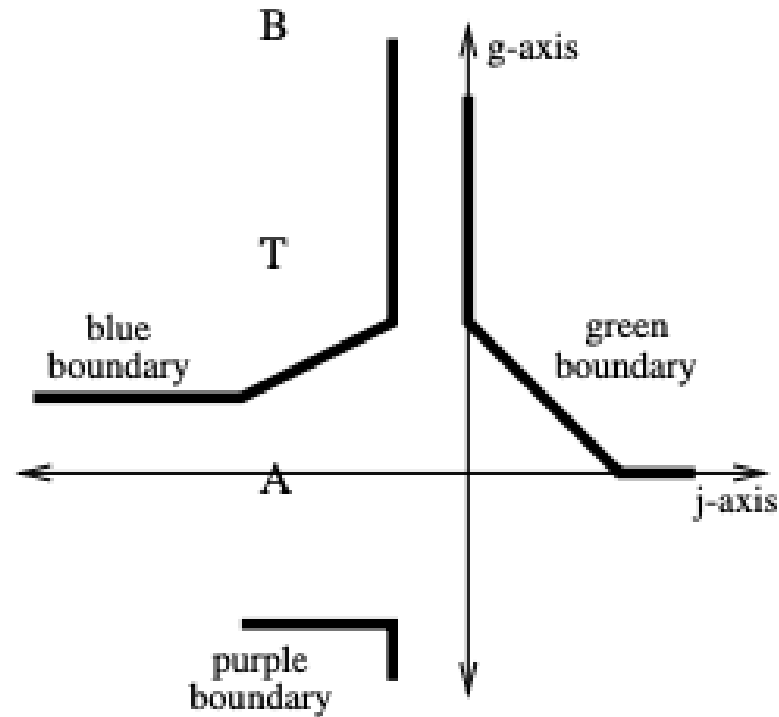
Linear Separation: the ability to linearly separate targets from non-targets in the color model being used



C. G. Healey, "Choosing effective colours for data visualization," *Proceedings of Seventh Annual IEEE Visualization '96*, San Francisco, CA, USA, 1996, pp. 263-270, doi: 10.1109/VISUAL.1996.568118.

Color for Visual Encoding

Color Category: the named color regions occupied by both the target and non-target elements



C. G. Healey, "Choosing effective colours for data visualization," *Proceedings of Seventh Annual IEEE Visualization '96*, San Francisco, CA, USA, 1996, pp. 263-270, doi: 10.1109/VISUAL.1996.568118.

Color for Visual Encoding

Another approach—Control:

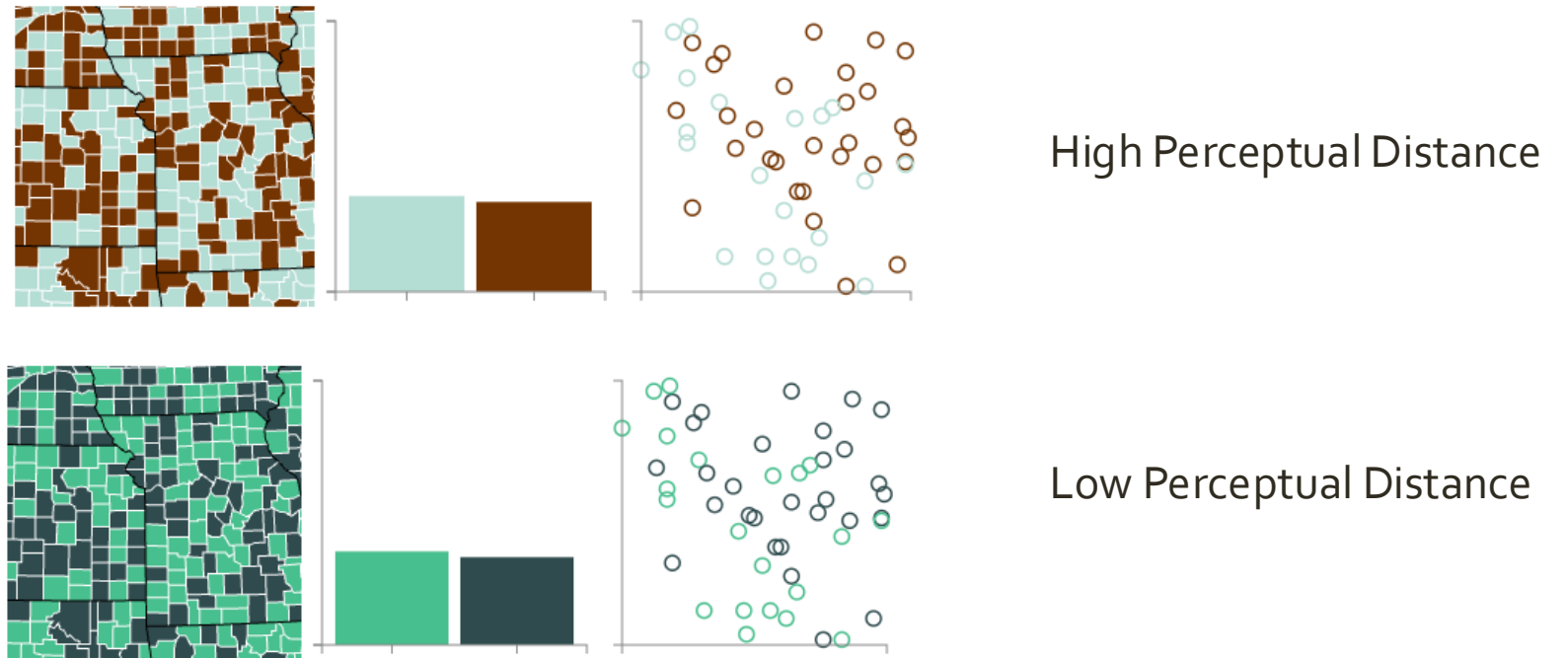
1. Perceptual Distance (CIEDE2000)
2. Name Difference
3. Pair Preference

The idea behind this approach is to prioritize aesthetically pleasing colormaps

C. C. Gramazio, D. H. Laidlaw and K. B. Schloss, "Colorgorical: Creating discriminable and preferable color palettes for information visualization," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 23, no. 1, pp. 521-530, Jan. 2017, doi: 10.1109/TVCG.2016.2598918.

Color for Visual Encoding

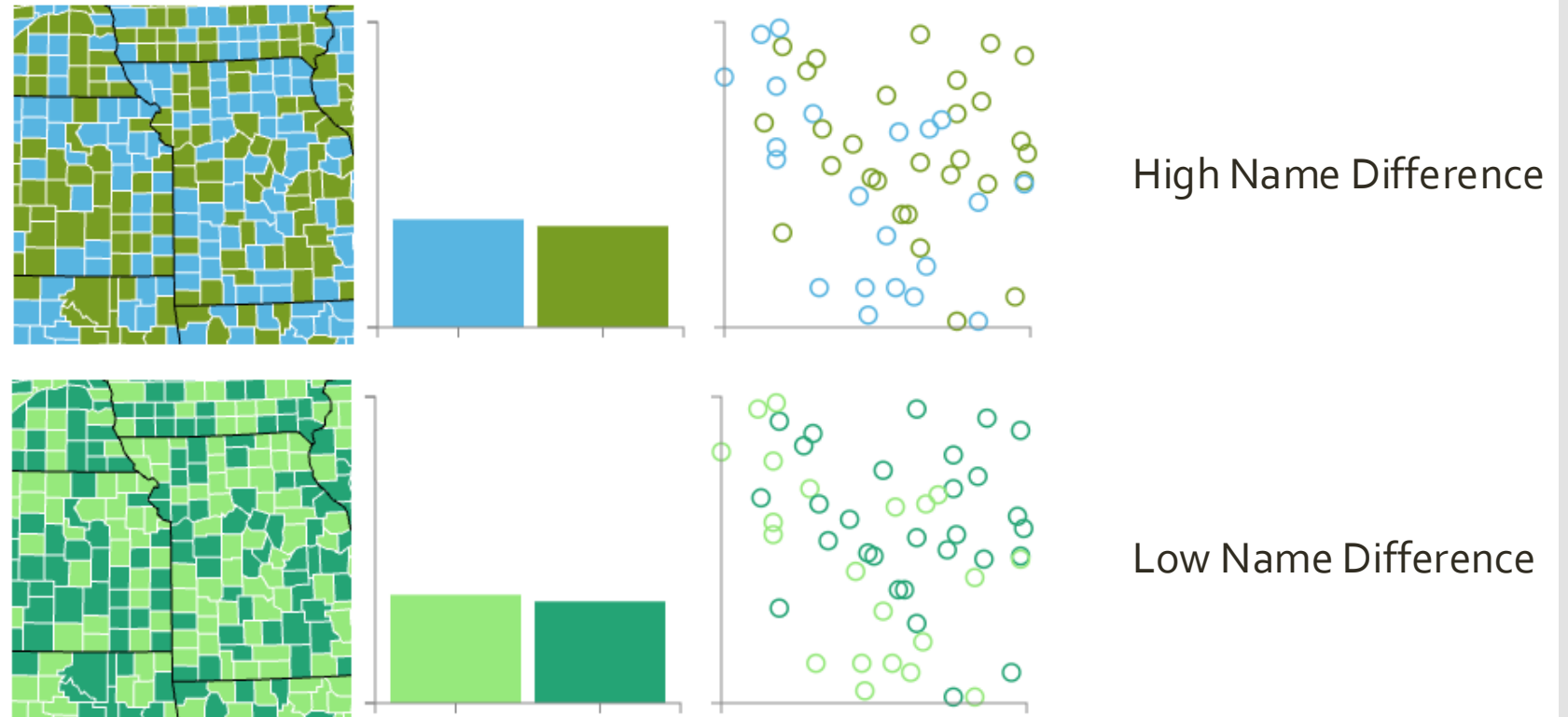
Perceptual Distance: A measure of how perceptually distinct two colors are. They use one based off CIEDE2000



C. C. Gramazio, D. H. Laidlaw and K. B. Schloss, "Colorgorical: Creating discriminable and preferable color palettes for information visualization," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 23, no. 1, pp. 521-530, Jan. 2017, doi: 10.1109/TVCG.2016.2598918.

Color for Visual Encoding

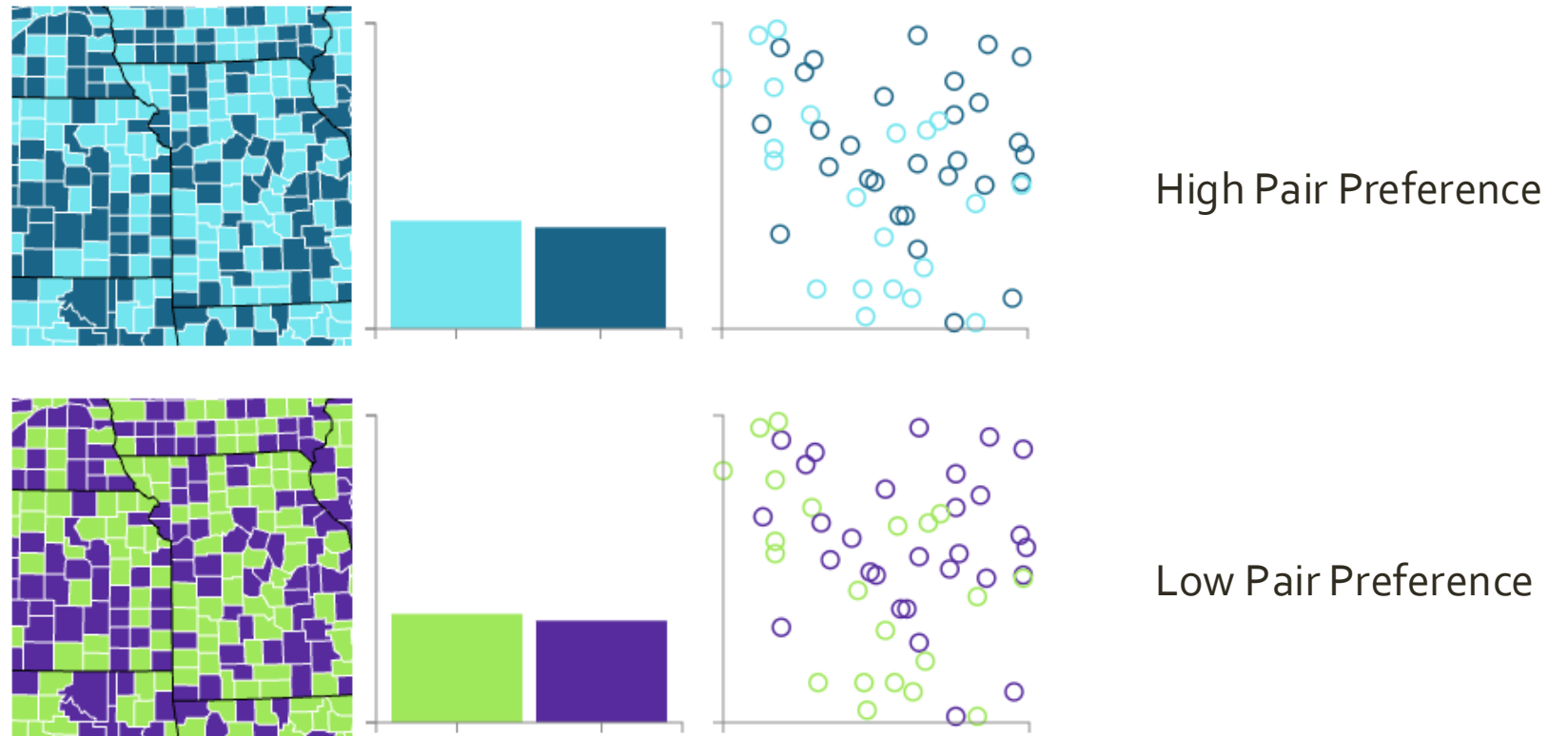
Name Difference: captures the degree to which two colors have distinct color-name association frequency distributions



C. C. Gramazio, D. H. Laidlaw and K. B. Schloss, "Colorgorical: Creating discriminable and preferable color palettes for information visualization," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 23, no. 1, pp. 521-530, Jan. 2017, doi: 10.1109/TVCG.2016.2598918.

Color for Visual Encoding

Pair Preference: reflects people's preference for color combinations that contain cool colors that differ in lightness and are similar in hue



C. C. Gramazio, D. H. Laidlaw and K. B. Schloss, "Colorgorical: Creating discriminable and preferable color palettes for information visualization," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 23, no. 1, pp. 521-530, Jan. 2017, doi: 10.1109/TVCG.2016.2598918.

Color for Visual Encoding

Another approach—Control:

1. Perceptual Distance (CIEDE2000)
2. Name Difference
3. Pair Preference

Try for yourself: <http://vrl.cs.brown.edu/color>

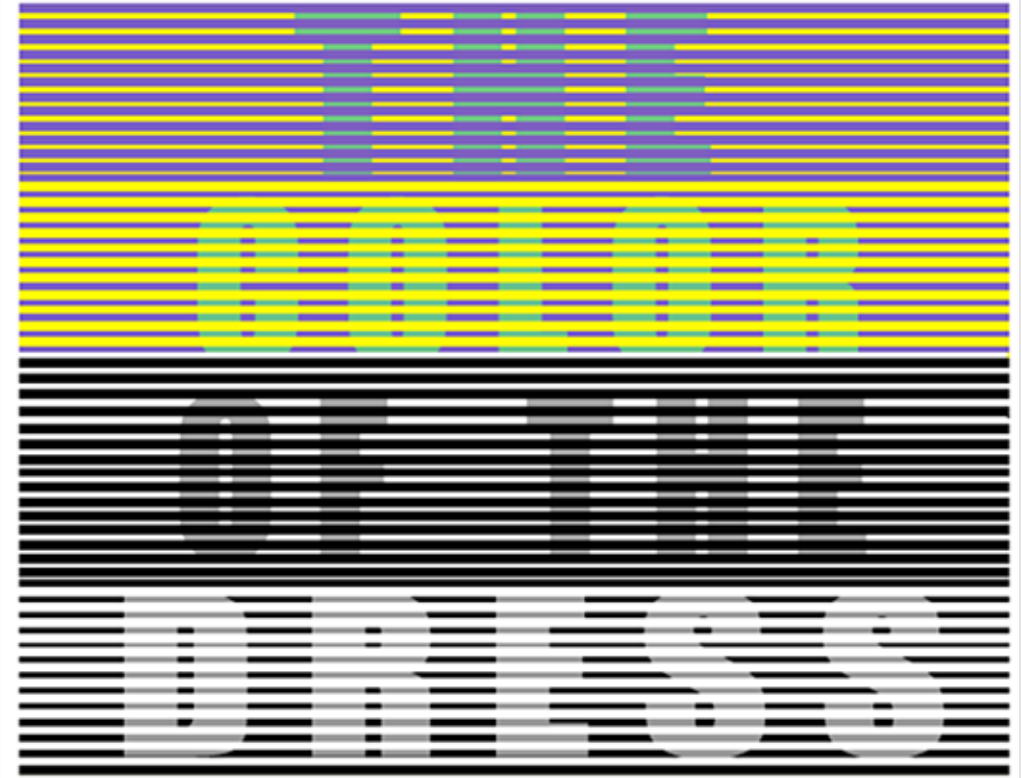
What do you notice?

C. C. Gramazio, D. H. Laidlaw and K. B. Schloss, "Colorgorical: Creating discriminable and preferable color palettes for information visualization," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 23, no. 1, pp. 521-530, Jan. 2017, doi: 10.1109/TVCG.2016.2598918.

Color for Visual Encoding

Which approach do you think is preferable / more effective? Why?

Color phenomena



Color phenomena

Caveat 1: color is perceived in context



Which small square is **darker green**?

R.B. Lotto, & D. Purves, An empirical explanation of color contrast, Proc. Natl. Acad. Sci. U.S.A. 97 (23) 12834-12839, <https://doi.org/10.1073/pnas.210369597> (2000).

Color phenomena

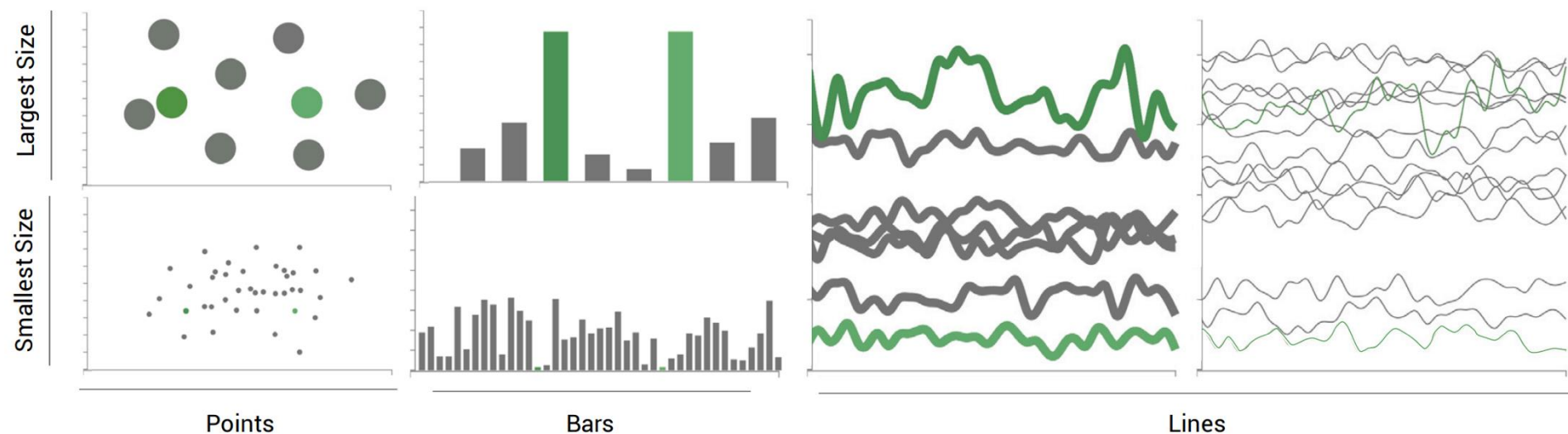
Caveat 2: difference is relative



R.B. Lotto, & D. Purves, An empirical explanation of color contrast, Proc. Natl. Acad. Sci. U.S.A. 97 (23) 12834-12839, <https://doi.org/10.1073/pnas.210369597> (2000).

Color phenomena

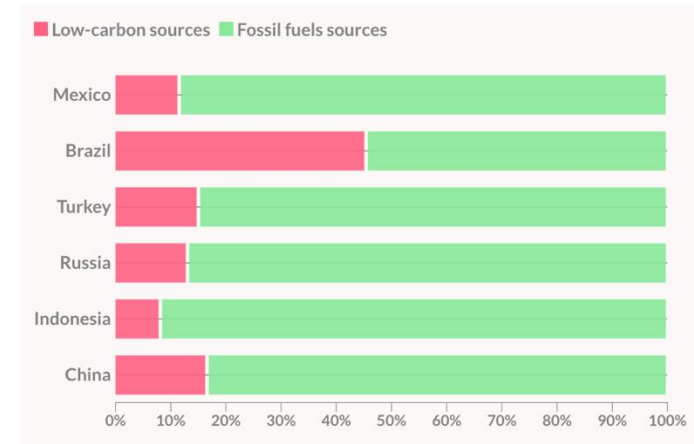
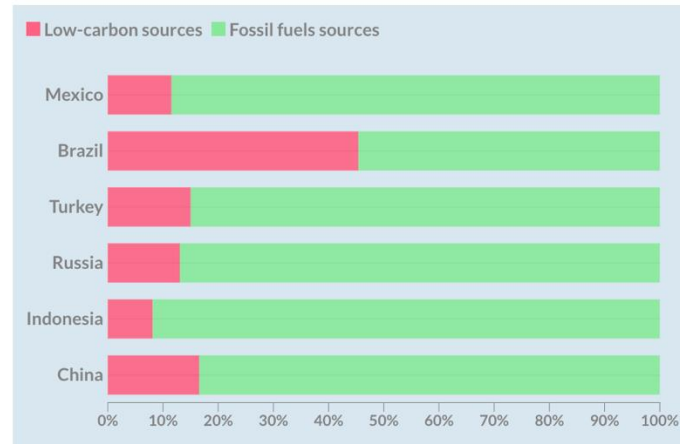
Caveat 3: Size of mark matters



D. A. Szafir, "Modeling Color Difference for Visualization Design," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 24, no. 1, pp. 392-401, Jan. 2018, doi: 10.1109/TVCG.2017.2744359

- Be careful when layering colors

Color phenomenon



❌ (large text only)



CONTRAST
L^c 50.5

✅ (for icons & graphics)



CONTRAST
L^c 38.5

❌



CONTRAST
L^c 0.0

✅



CONTRAST
L^c 62.2

✅ (for icons & graphics)



CONTRAST
L^c 50.3

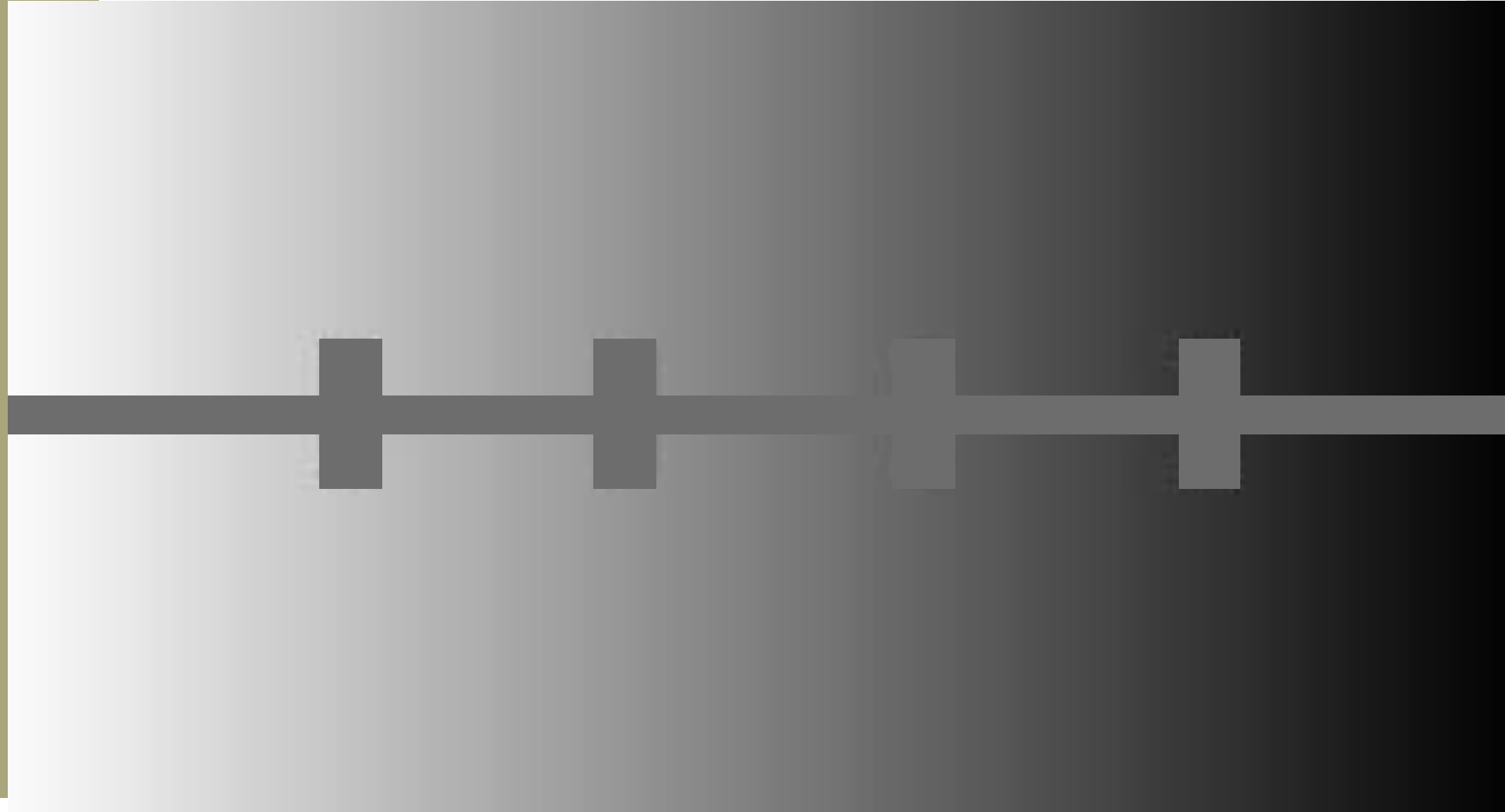
✅ (for icons & graphics)



CONTRAST
L^c 18.8

Caveat 2a: so are brightness and contrast

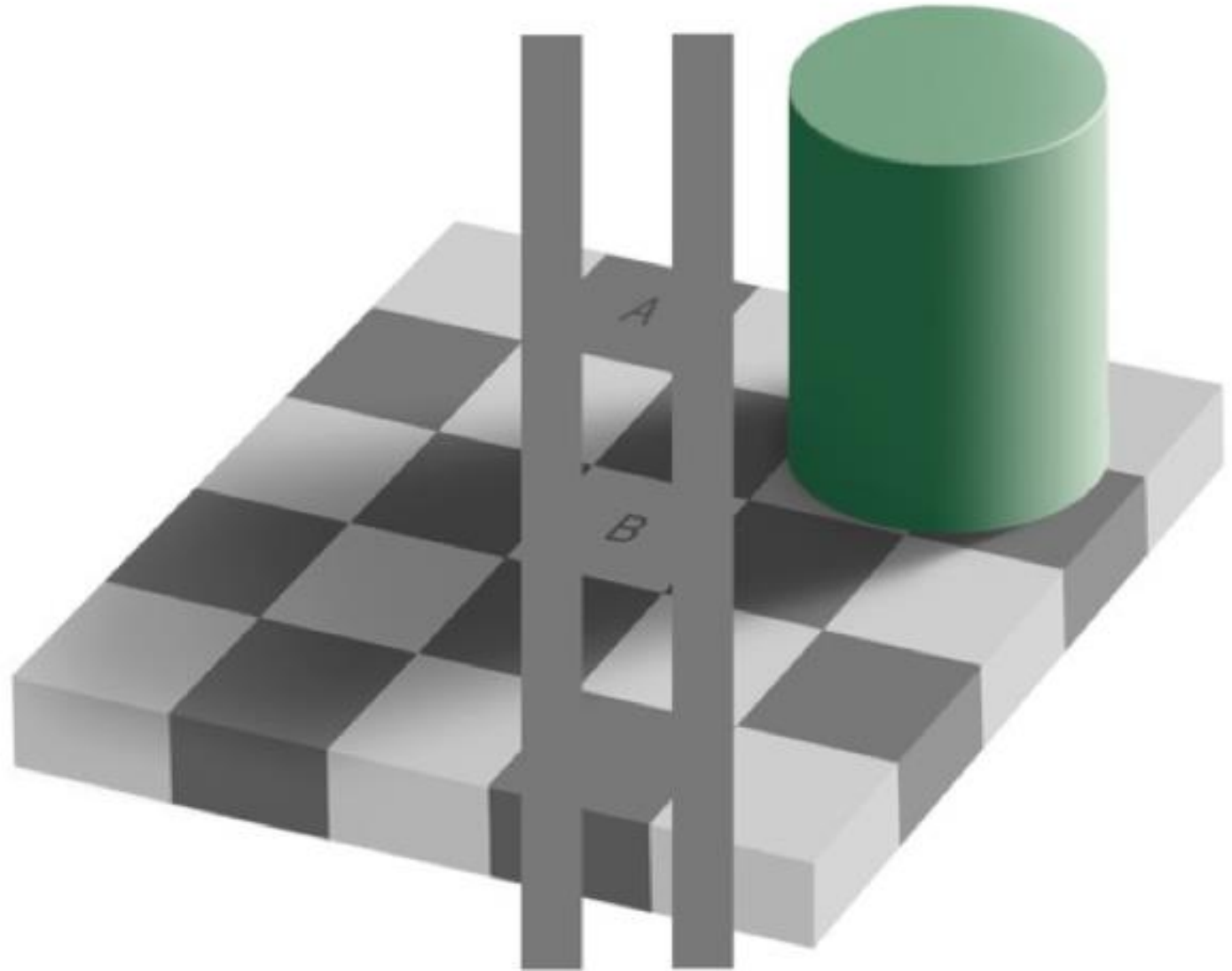
Color
phenomena



Ware, Colin. (1988). Color sequences for univariate maps: theory, experiments and principles. IEEE Computer Graphics and Applications. 8. 41-49. [10.1109/38.7760](https://doi.org/10.1109/38.7760).

Caveat 3: mental models > perception

Color
phenomena



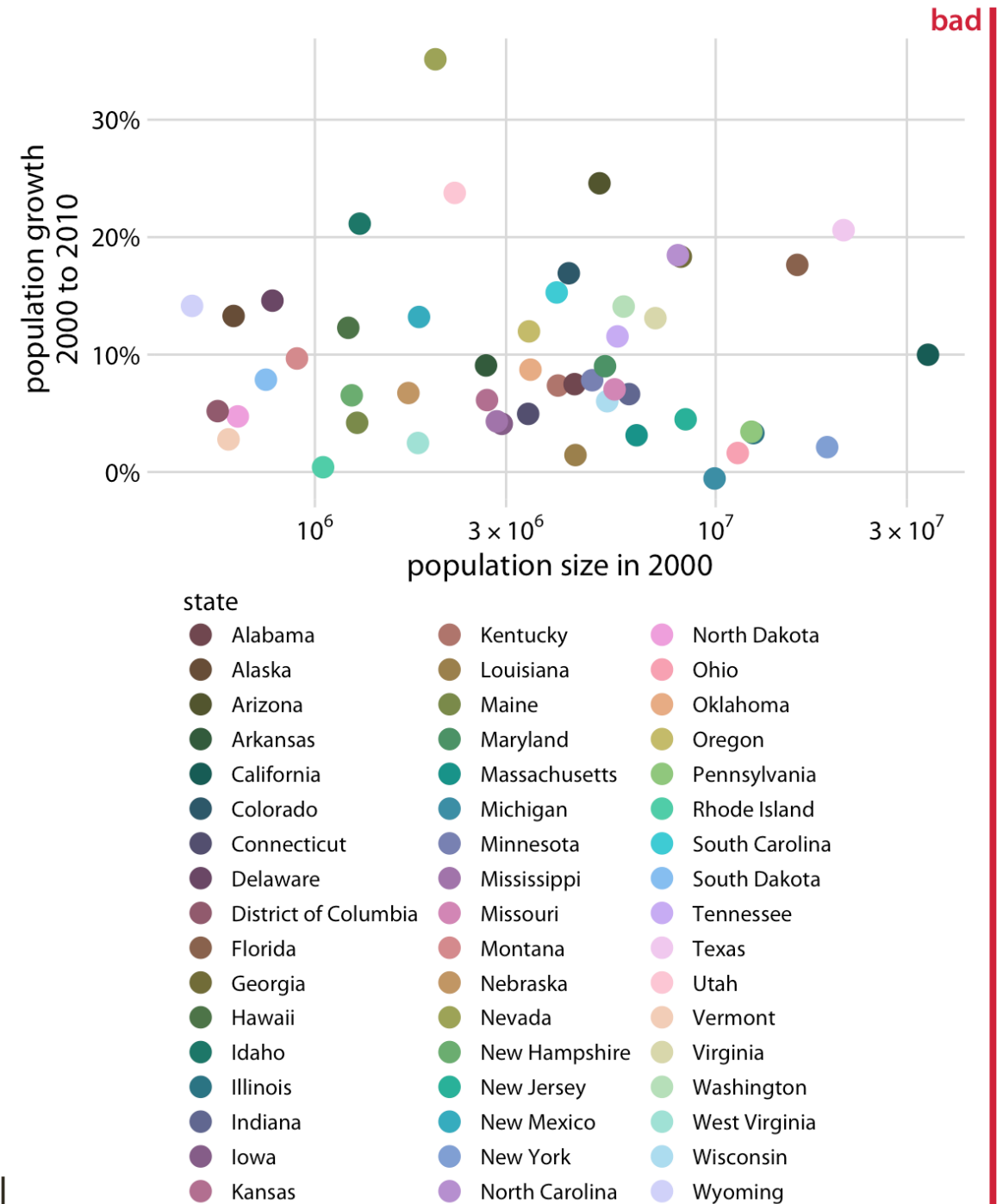
Adelson, Edward H. 1995. "Checkers shadow Illusion."

Color Problems

- Using a poor color scheme can also cause issues with your visualization

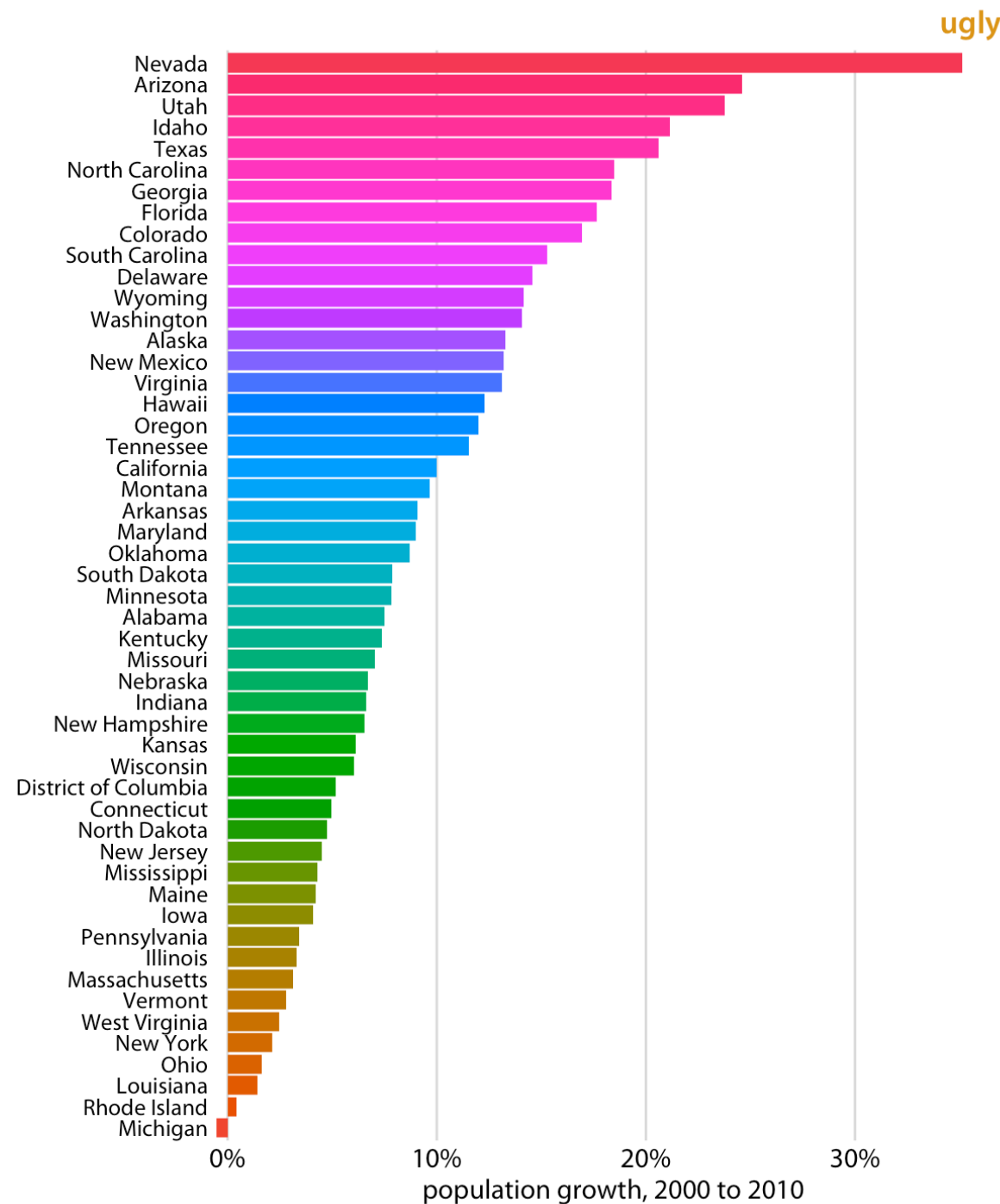
Color Problems

- Hue mapped to too many categories



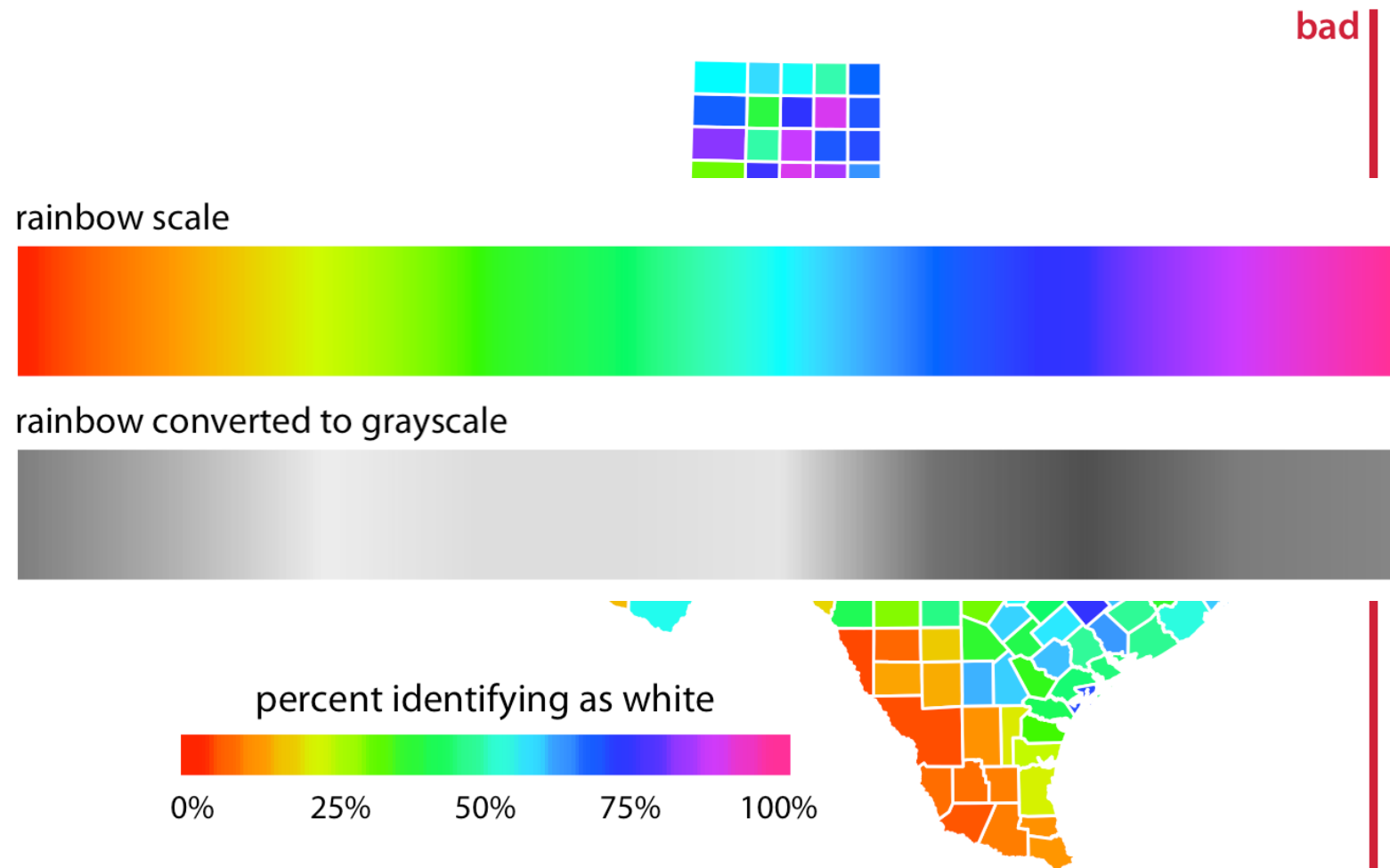
Color Problems

- Hue not mapped to data



Color Problems

- Rainbow obscures real patterns (it is not monotonic)



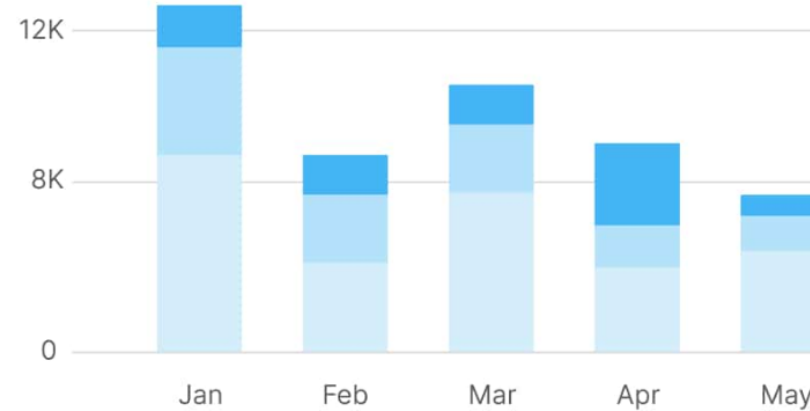
Color Problems

- Similar colors are hard to differentiate

✕ Similar colors

Product Sales

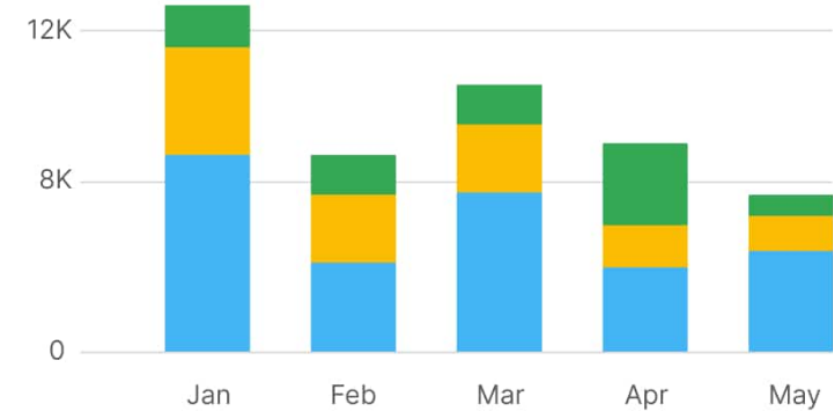
□ Clothing □ Equipment ■ Accessories



✓ Different colors

Product Sales

■ Clothing ■ Equipment ■ Accessories



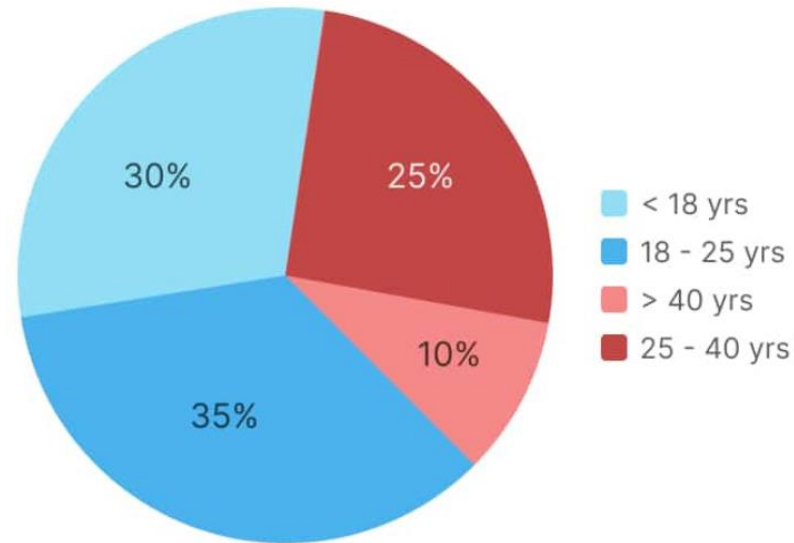
<https://matthewstrom.com/writing/how-to-pick-the-least-wrong-colors/>

Color Problems

- Similar colors imply relationships

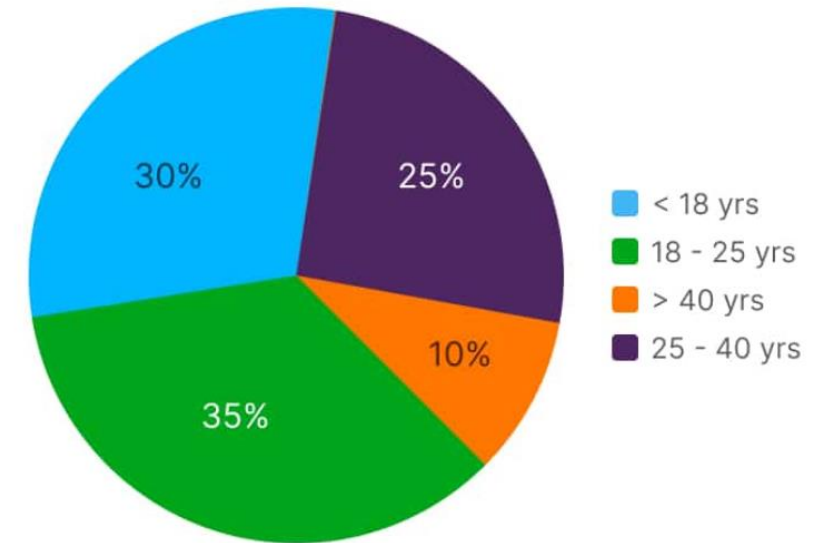
✗ Uneven color distribution

Age distribution of users



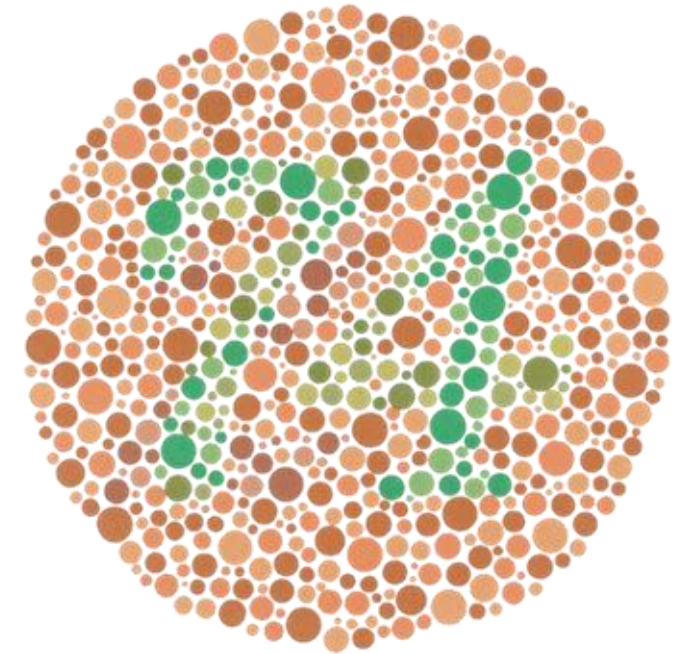
✓ Even color distribution

Age distribution of users



<https://matthewstrom.com/writing/how-to-pick-the-least-wrong-colors/>

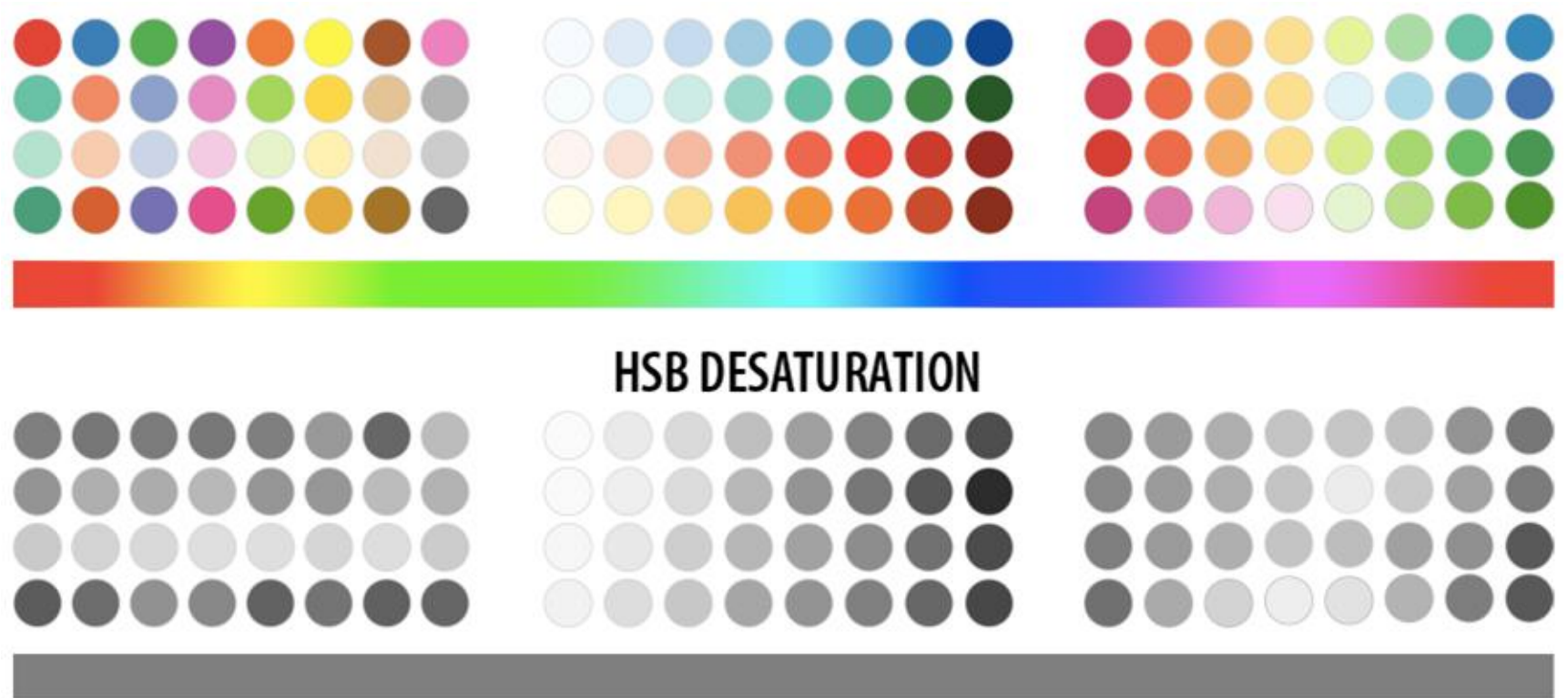
“colorblindness”



1 out of every 8 people has just 2 types of color receptors (rather than 3)

What happens
when you
print?

- Need color scheme that converts well to grey scale



Colorbrewer palettes

- colorbrewer.org provides a whole bunch of palettes that can help us avoid these issues
- This makes life a lot easier for us!

QUALITATIVE

set1



set2



pastel2



dark2



SEQUENTIAL

blues



greens



reds



ylorbr



DIVERGING

spectral



rdylbu



rdylgn



piyg



Colorbrewer palettes

QUALITATIVE

set1



set2



pastel2



dark2

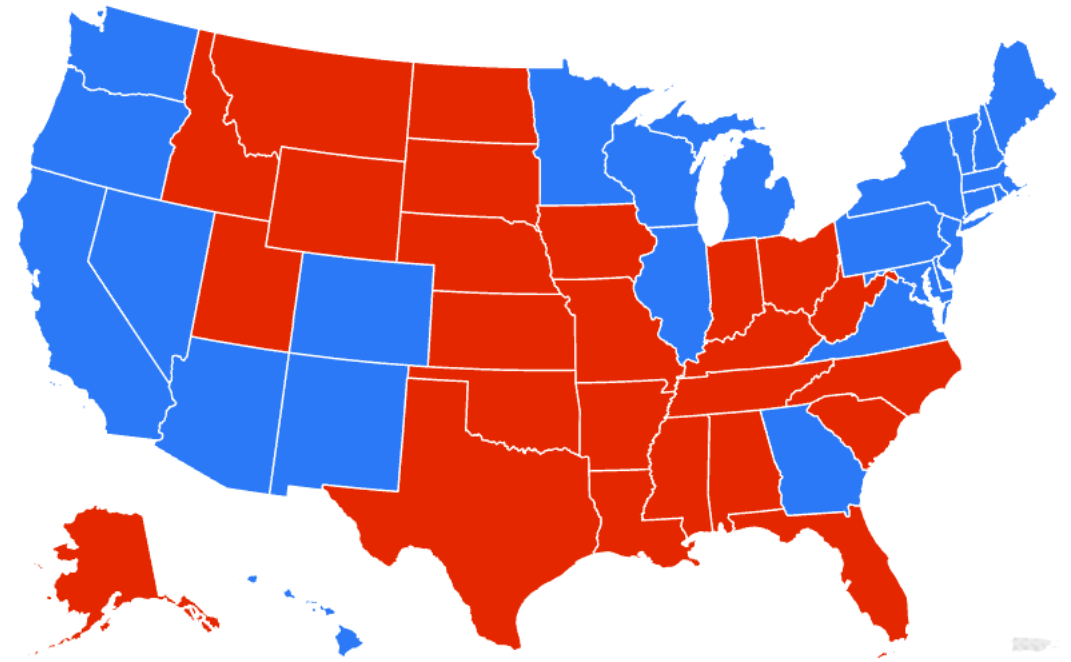


Good for nominal data

Categorical color palette

2020 Presidential election: winner per state

■ Democrat ■ Republican



Source: [The Cook Political Report](#) • Created with the Flourish [Projection Map](#) template

Colorbrewer palettes

SEQUENTIAL

blues



greens



reds



ylorbr

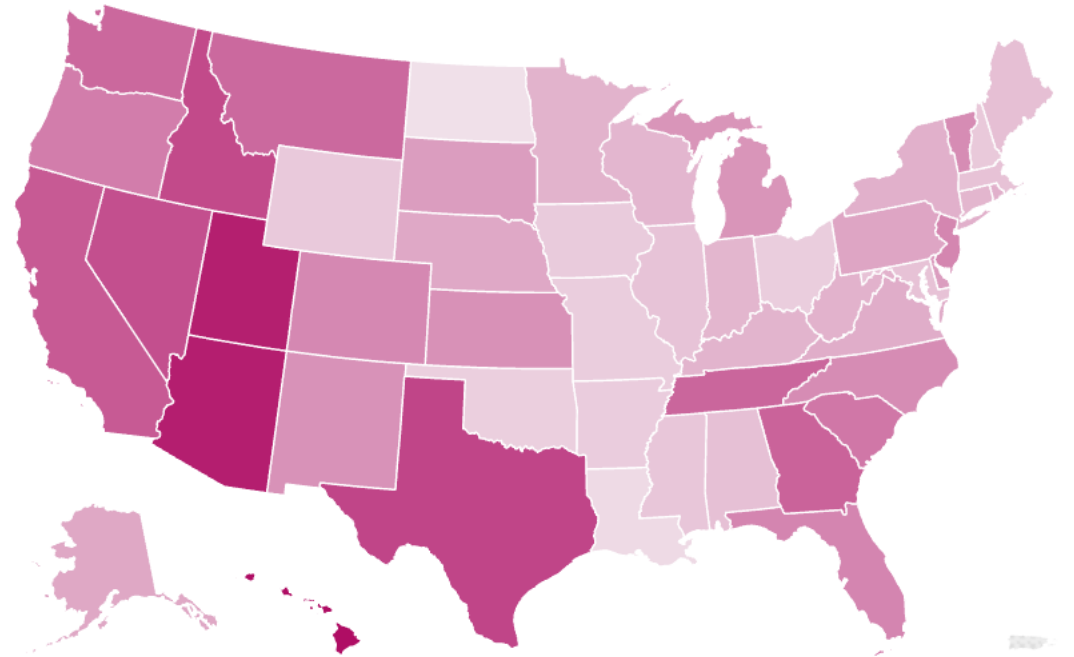


Good for sequential ordinal
& quantitative data

Sequential color palette

2020 Presidential election: vote change (%)

5.1  33.9



Source: [The Cook Political Report](#) • Created with the Flourish [Projection Map](#) template

Colorbrewer palettes

DIVERGING

spectral



rdylbu



rdylgn



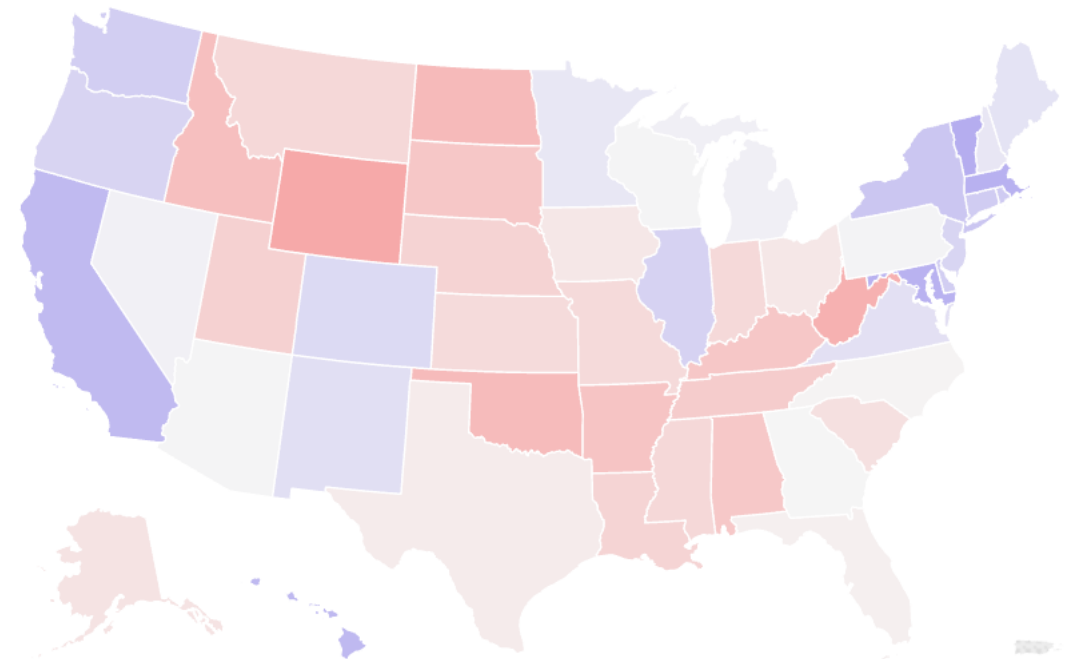
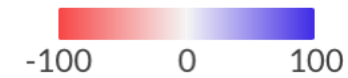
piyg



Good for diverging ordinal
& quantitative data

Diverging color palette

2020 Presidential election: margin of victory (%)



Source: [The Cook Political Report](#) • Created with the Flourish [Projection Map](#) template

Takeaways: Perception

- Visualization is about more than just aesthetics
- There are compelling **cognitive reasons** why some visualization techniques are helpful and others aren't
- The choices we make about **visual mappings** can have a significant effect on performance

Your turn!

- Work with a classmate or 2
- Find a two visualizations from trusted sources that visually encode data via color. One should use hue and the other saturation.
- Assess the use of color:
 - Is it perceptually accurate and easy to read?
 - Is it aesthetically pleasing?
 - Are there any optical illusions?
 - What would you change if you were to re-design the visualization?